Lesson Notes

MODULE 3 | LESSON 4

Probabilistic Scenario Optimization and Probabilistic Approaches Lesson Notes

Reading Time	30 minutes
Prior Knowledge	Students are expected to be familiar with the concepts developed in the previous lessons. In particular a solid grasp of the traditional mean-variance optimization is a necessary requirement.
Keywords	Probability Scenario Optimization (PSO), probabilistic risk measures, algorithms for these situations.

In the previous model we completed the coverage of the Black-Litterman model analysis and a few abuses of that model. Practitioners like to simplify things and, arguably, rightly so, but often actions have consequences. In this lesson, we will cover a completely different topic that, not unlike the Black-Litterman model, can be used to extend the validity and the usefulness of Markowitz's approach.

In the next lesson that starts a new module of the course, we will consider a very important problem: in all models we have visited so far it has been assumed that investors or economic agents have a clear view of what they desire (risk takers, risk averse, etc.) and that they always know exactly how to optimize their choices based on some type of utility functions. Is that always the case? The field of behavioral economics and finance would beg to differ and would say that the fully rational man of classic economics is just a simplification.

1. Probabilistic Scenario Optimization (PSO)

Imagine you are assigned to manage the investment portfolio of a company's pension plan. In particular, let's assume the pension plan is a defined benefit pension plan. In a defined benefit pension plan, the employers promise to pay a defined benefit to employees for life after they retire. In contrast, in a defined contribution plan (401(k), for instance) employees put their money in an employer-sponsored investment program, but the company is not responsible for making periodical payments.

As a portfolio manager for the pension fund, you may still have the long-term goal to maximize the risk-return profile of the portfolio. However, you need more than that. Depending on certain demographic and actuarial characteristics of the company's work-force (past and present), you may have a need to produce specific streams of income to be able to make the payments while insuring the long-term goal of optimizing the portfolio's risk-return profile. Or since the periodic payments may be indexed to inflation, how should you invest to make up for the higher and higher payments?

A similar issue would arise if you were asked to manage the investment portfolio for a property and casualty insurance company. There is no certainty about how much or how often the company has to make payments; however, a certain amount of money must be very liquid in case it had to make large payments to the clients it has insured. Generally speaking, these companies invest with a short horizon as their liabilities have short duration.

The problem does not change much if we consider life insurance companies. Insurance companies sell policies that promise a payment to the policyholders if an event covered under the policy occurs during the life of the policy. The funds received from the policyholders (premiums) are invested until needed for payout. Traditionally, life insurance companies worry the most about a mismatch in the duration of assets and liabilities (as their liabilities are overall long term), cash flow volatility, and credit risk. Thus, to reduce the first type of risk, they focus on a minimum return target (usually set equal to the actuarial assumed rate of growth of policyholder reserves, since it is the minimum return needed to meet payouts), and tend to invest mostly in fixed income assets. Often, they could face legal and regulatory constraints as well.

These issues do exist even if we were managing the investment portfolio for a foundation or an endowment where the goal is to grow the capital after the spending requirements are met but not to risk it in any unreasonable manner. In some cases there could be additional legal and regulatory constraints to consider.

Where are we going with all this? We hope it helps to illustrate the point that the objective function to maximize cannot always be a simple mean-variance representation and that additional elements should be added to it. And, furthermore, we may have a view about several possible scenarios that may develop during the relevant period under consideration and that could affect different securities and/or different classes of assets in different manners.

We need to accept that a generic allocation $w^T = (w_1, w_2, \ldots, w_n)$ is not just a vector that represents the amount of units or the percentage of wealth invested in each of the n securities in a given portfolio. The allocation must be considered as the result of a decision process together with the available information. And, if the available information had been different, we would have ended up with a different allocation vector $\tilde{w}^T$ for our portfolio.

If we wanted to evaluate the pros and cons of a specific allocation, we would need to evaluate the decision process behind it. It is really quite similar to what is done in statistics when we introduce many estimators and then study the properties of such estimators from a probabilistic point of view. This is arguably a good description of PSO.

It must be stressed from the start that PSO is not a simple task and that it is part arts and part science. In its essence, PSO is a risk-based technique aimed at facilitating the implementation of goal-based investing in portfolios managed by or for specific institutions. The decision making extends to long-term investments that investors can optimize and that do not include just the max

return for a given level of risk. Putting it in a more simplistic manner and simplifying to the maximum extent, we could try to decide which of two allocations would be most suitable to us given that we foresee certain potential scenarios and are in a position to assign probabilities to each of the scenarios. We could run many simulations and consider which of the two allocations of investments minimizes the probability to be below a certain threshold or we could mix this probability to reduce the need to minimize the risk, etc.

1.1. How Is It Done?

This is not a simple task, and there is more than one way to implement it, but in the end, it is based on an apparatus of well-conceived simulations. Lots of simulations.

Here, we provide a description of the steps involved.

Step 1: Fix the investment horizon. This could be a period of, say 10 years. Such period can be split in even subintervals or we could use an uneven time step discrete grid with higher frequency in the short terms than in the long terms, say the time period could be 1 week for the first 6 months, 1 month for the next 1.5 years, 6 months after that, etc. These will be the steps at each of which we will perform a Monte Carlo simulation.

Step 2: Select the investment universe, which is obviously no surprise, and indicate the allocation constraints (no short selling, max currency exposure, max single asset exposure, etc.). Furthermore, we must specify the investment risk-return profile for our client. For instance:

- We could have a risk-averse profile and set the target return similar to risk-free rate. Say we want to achieve a return of 1% per year. In this profile, we could set a risk adequacy so that our capital never loses value starting from a specific period, say the third year or the fifth.
- Alternatively, we could have a more accommodating risk-return profile and set a target return of, say LIBOR + 200 bps. We could also allow for some capital risk over the short term and enforce tighter capital insurance in the medium to long terms.
- Or, we could adopt a risk-return profile more tolerant of risk in which the return goal could be set to, say, 5% a year at least and no control on risk over the full time horizon of the investment.

Step 3: Decide on the type of optimization: multi-period (usually) or single-period. In the case of a single period optimization, say 5 years, one would not model any intermediate risk. In the multi-period, one can set liquidity constraints for a certain number of years, i.e., during this period, we periodically check the adequacy constraints at the end of every year and after that period, we no longer check for portfolio weights' adequacy.

Step 4: We need to specify an objective function, which could be the probability or weighted probability of a total return higher than our minimum goal while complying with allocation constraints and risk limits. Regarding the risk limits, the risk measure to be used could be implemented via a one-tail VaR.

Step 5: Before running the Monte Carlo simulations, we have to set up the scenarios in terms of probability distributions. A multivariate normal is a common choice, but it is not the only possible choice. Furthermore, one could also think of introducing a few ad-hoc views that will serve to stress test our results. We could imagine a booming economy for the first 5 years, which could be followed by a sharp decline or, vice versa, an economy that starts with a sharp decline in the equity markets followed by a protracted expansion until another threshold time and then contracts again. Notice

that we need to generate Monte Carlo scenarios on a variety of risk factors, which could include market indices, individual securities, currency exchange rates, and credit curves for government and credit issuers. This requires that we collect time series data over a pre-specified period and then compute daily returns, correlations, etc.

Step 6: Generate the entire quasi-random space of admissible asset allocations that satisfy the asset allocation constraints given above. This must be understood in the sense that, even if we start with asset allocations that fulfill certain requirement of no short selling, or limit on the concentration, under certain scenarios, as the value of the assets change, we could break through those constraints, and therefore, we need to re-check them. Recall that the weights are in market capitalization percentage of the total portfolio.

Step 7. Reduce the admissible asset allocations to the set of risk-adequate portfolios with respect to the investor's risk-profile. In other words, we filter out those portfolios that do not comply with our ambitions.

While this step could have been performed on the space of all admissible asset allocations, working on the sample allocations that are compliant with the allocation constraints is advantageous from a computational point of view.

This step could take the form of assigning boundary conditions, like that we do not want to lose more than a certain amount with a certain probability, i.e., we want the α quantile profile of the investor to be higher than such threshold. Or it would be a constraint where the right quantile of the optimal portfolio at any selected time point t has a probability $1 - \alpha$ to be greater than or equal to the investor's return required return. In this way, if we choose a low α level, we are imposing a risk cap. By choosing a medium to high α-level, we can impose a probabilistic return floor.

Step 8: Measure the probability required and optimize the objective function. This can be quite different than what we are accustomed to. In fact, we would not just look for the maximum return, which could be achieved by different portfolios under different scenarios in general. But, we could maximize the probability of achieving a minimum time dependent target return level, say l_t^A where $t \in T$ and T is the universe of time steps. Indicating with Θ the set of portfolios that made it to this point, we are trying to determine the portfolio(s) such that:

$$\max_{\theta \in \Theta} \left\{ \sum_{t \in T} k_t P[R_\theta > l_t^A] \right\}$$

where R_θ represents the return on portfolio θ, $k_t \in K$ is a multi-period weighting scheme that allocates a positive weight to each $t \in T$. If we assigned full weight to only a discrete set of points $t \in T$, then the other points would be irrelevant. In an extreme case, we could attach the entire weight to one point only, say the last point.

Step 9: Measure the performance and compare portfolios, i.e., track the investment performance over time and compute the probability to achieve or beat a target return across time or the probability to fall short of it.

1.2 Low Discrepancy of Quasi-Random Numbers

When running Monte Carlo simulations, admissible asset allocations can be a significantly large set, running into the millions or billions. As if this were not enough, the underlying space depends on

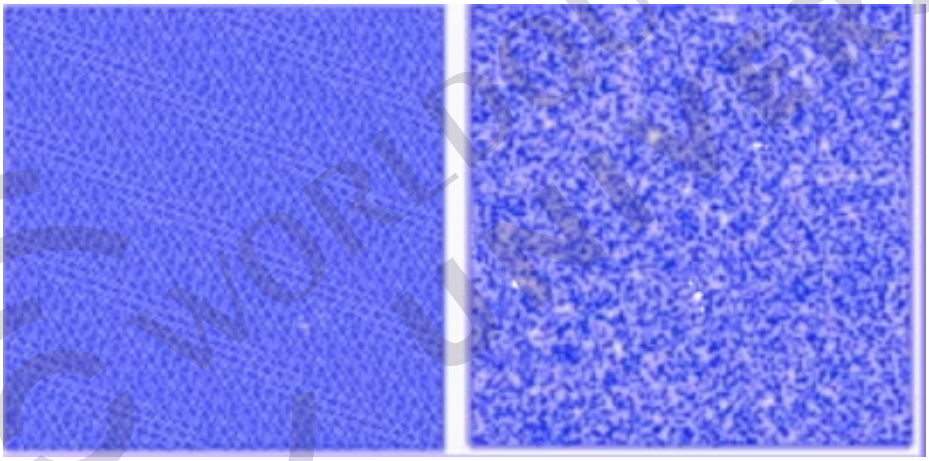
multidimensional sets of variables. For instance, if we model 100 assets and need to compute a probability (an area), this is like doing integration on a high dimensional space. Evaluation of high-dimensional integrals is performed by stochastic Monte Carlo simulations. Simulations that use **quasi-random numbers** can achieve faster convergence and better accuracy than simulations via conventional pseudo-random numbers.

The purpose of **low discrepancy methods** is to build a series of simulations by sampling from the interval $[0, 1]$ in such a way that every draw is further away from the previous one, and the draws are as far apart from each other as possible to guarantee better overall coverage and avoid situations in which parts of the domain contain many points and others are nearly deserted.

Monte Carlo integration uses a finite quantity of pseudo-random numbers. A critical issue with these numbers is that they may not be perfectly equi-distributed, i.e., the finite quantity of pseudo-random numbers used are not spread uniformly throughout the domain of the integrand function, which leads to poorer results. Equi-distribution, and therefore accuracy, improve as more random numbers are used, but this requires longer run-times. Low-discrepancy sequences that are well-distributed even for small quantities were introduced.

Figure 1 compares 10,000 simulations on the square $[0, 1) \times [0, 1)$ using traditional methods (right) and quasi-random numbers (left).

Figure 1: Comparing Simulations using Quasi-Random Numbers and the Traditional Technique



To understand the concept of discrepancy, let's consider a hypercube like $I_s = [0, 1)^s$ for some s an integer greater or equal to 1. Consider now M points in $I_s : z_1, z_2, \ldots, z_M$. Furthermore, consider a subset J or I_s, i.e.$I_s \subset J$ and let $A(J)$ be the number of points $z_1, \ldots, z_M$ that are in J. Finally, let $V(J)$ denote the volume of J. Then, the discrepancy $D(J, M)$ is defined as

$$D(J, M) = |A(J)/M - V(J)|.$$

Intuitively, the discrepancy is the difference between the proportion of points in J compared to the volume of the box J.

We can also identify a worst-case discrepancy, i.e., the worst-case distribution of a sequence of points $z_1, z_2, \ldots, z_M$, such that

$$D^{\times}(M) \equiv \max_J D(J, M).$$

The goal of a low-discrepancy sequence is to minimize $D^{\times}(M)$.

One important method to generate low-discrepancy sequences is provided by Halton sequences. In this method, each dimension is represented in a different prime base $(e.g., 2, 3, 5, 7, \ldots)$. To generate the n-th point in a sequence, consider the b-ary expansion of a number n $\sum_{i=0}^{\infty} a_i b^i$ where $a_i \in \{0, 1, 2, \ldots, b-1\}$. The n-th Halton point $H(n)$ is the radical-inverse function in base b :

$$H(n) = \sum_{i=0}^{\infty} a_i b^{-1-i}.$$

Example. Consider the base $b = 2$. And let's draw the first 7 numbers $H(1), \ldots, H(7)$. To do this, we need to write each number in its $b = 2$-ary expansion:

$$1 = 1, 2 = 10, 3 = 11, 4 = 100, 5 = 101, 6 = 110, 7 = 111.$$

Now, pay attention to the fact that in representing 2 for instance, we have $a_0 = 0$ and $a_1 = 1$. In representing 4, $a_0 = 0, a_1 = 0, a_2 = 1$, etc. Then

$$H(1) = 1 \times 2^{-1} = 0.1$$

which, in decimal notation, is $1/2$. Then

$$H(2) = 0 \times 2^{-1} + 1 \times 2^{-2} = 0.01,$$

or, in decimal notation,$H(2) = 1/4$. Moving along, we find also:

$$H(3) = 1 \times 2^{-1} + 1 \times 2^{-2} = 0.11,$$

i.e.,$3/4$ in decimal notation.

$$\ldots\ldots$$

$$H(7) = 1 \times 2^{-1} + 1 \times 2^{-2} + 1 \times 2^{-3} = 0.111$$

or $7/8$ in decimal notation. As one can see, we are slowly populating the interval between 0 and 1.

How do we generate sequences in $(0, 1) \times (0, 1)$ or higher dimensions? We partition the first $(0, 1)$ interval using a 2-ary expansion as we did above. For the second space $(0, 1)$, we use instead a 3-ary expansion. Let's call these numbers using $\tilde{H}$. Hence, we start with

$$1_{10} = 1_3, 2_{10} = 2_3, 3_{10} = 10_3, 4_{10} = 11_3, 5_{10} = 12_3, \ldots,$$

where the symbols a_{10} and a_3 indicate in which base the number a is written. Then, we use the same approach as before, i.e.,

$$\tilde{H}(1) = 0.1 = 1 \times 3^{-1} = 1/3, \tilde{H}(2) = 2 \times 3^{-1} = 2/3, \tilde{H}(3) = 0 \times 3^{-1} + 1 \times 3^{-2} = 0.01 = 1/9, \tilde{H}(4) = 4/9,$$

and so on. Finally, we consider the pairs of numbers:

$$(1/2, 1/3), (1/4, 2/3), (3/4, 1/9), \ldots$$

Students interested in this topic could read the paper Monte Carlo integration with quasi random numbers which contains also relevant Python code.

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1.3 Ordering, Lexicographical Ordering

Admissible portfolios must be ordered so that we can apply to each its Halton sequence. There is a bit of a problem in doing this because say that the universe of investments consists of only four assets, A, B, C, and D, and the tick sizes (weight of the assets in the portfolio) can only be one of the following values: $\{0, 0.25, 0.50, 0.75, 1\}$. Which portfolio comes first among $(0.25, 0.25, 0.5, 0)$ and $(0.25, 0.50, 0, 0.25)$ if we were to list them? To do this we need to introduce some suitable ordering. In this case, **lexicographical ordering** can be used (but it is not the only one possible). To this purpose, we rewrite the list $\{0, 0.25, 0.5, 0.75, 1\}$ as $\{a, b, c, d, e\}$. Now, we make further assumptions that we are given minimum and maximum allocations for each asset, say

$$A : [0.25, 0.50], B : [0.25, 0.75], C : [0, 0.50], D : [0, 0.75]$$

and we also have the usual constraint that the sum of the weights is equal to 1.

We can represent these constraints based on our identification above also in the format:

$$A : [1, 2], B : [1, 3], C : [0, 2], D : [0, 3].$$

We can build an ordered set of allocations by starting to allocate the minimum amount allowed for each asset in the alphabetical order where $0 < 1 < 2 < 3 < 4$ but *thought of as letters and not numbers*!

So:

$$\text{Allocation } 1 : 0.25, 0.25, 0, 0.5 = \{b, b, a, c\};$$

$$\text{Allocation } 2 : 0.25, 0.25, 0.25, 0.25 = \{b, b, b, b\};$$

$$\text{Allocation } 3 : 0.25, 0.25, 0.5, 0.0 = \{b, b, c, a\};$$

and so on. You should be able to see the lexicographical order of the letters starting from the first to the last as if looking for "words" in a dictionary. The words are "bbac," "bbbb," "bbca," etc. How many allocations are there? It is possible to find that number, but the task requires a complex recursive formula that depends on several inputs and as this is just an introductory lesson to the topic, we will not pursue it.

However, once the notion of lexicographical order is understood, we can use the portfolios in a sequence that can be programmed.

2. Portfolio Optimization Using a Probabilistic Risk Measure

Earlier, we mentioned that an optimal portfolio cannot always be expressed in terms of mean and variance. There are instances where an investor cannot measure the risk of their portfolio only based on volatility alone. In Module 1, Lesson 1, we mentioned that semi-variance can also be used, but other risk measures have been suggested as well, like the mean absolute deviation for instance. Further, in Module 2, we looked at how to include skewness and kurtosis in an optimization scheme. We should not forget VaR as another possible choice, despite the fact that it suffers from certain known limitations. Minimax measures have also been suggested: We compute the max of certain risks that are relevant for our situation and then choose the portfolio that minimize the maximum risk.

This is less general than working with probabilistic scenarios, but it is still a pace in a direction that is outside the classical optimization problems in portfolio theory. The aim of this section is to introduce students to a different way to state the problem.

Let's look a little more at how this could be done. To this purpose, we assume that W_0 is the investor's initial wealth, which they would like to invest in a universe of N possible risky assets, $S_1, \ldots, S_N$. To avoid confusion with wealth, we use the symbols $x_1, \ldots, x_N$ to denote the percentage of W_0 invested in each asset. The usual constraints apply, i.e., the space of viable portfolios is defined as:

$$\mathcal{X} = \{(x_1, \ldots, x_N) : \sum_{i=1}^{N} x_i = 1 \quad \text{and} \quad x_i \geq 0\}$$

Now, let R_i be the return on asset i for $i = 1, 2, \ldots, N$. Furthermore, we assume that the R_i's are stochastic, and, for convenience, that

$$R_i \sim N(\mu_i \sigma_i^2).$$

We do not know μ_i, but it can be estimated if we have observations for T periods, that is

$$\hat{\mu}_i \equiv r_i = \frac{1}{T} \sum_{t=1}^{T} R_{it}$$

where R_{it} simply denotes the return for the i-th asset during the t-th time period. Then, we can estimate the return on the portfolio as

$$r(x_1, x_2, \ldots, x_N) \sum_{i=1}^{N} r_i x_i.$$

If the σ_is are not known, they can be estimated as well.

So far, nothing is really different from what we have done many times already. However, now we introduce what we call a **probabilistic risk measure**:

$$\omega_p(x_1, \ldots, x_N) = \min_{1 \leq i \leq N} \{P[|R_i - r_i| x_i \leq \theta \cdot \epsilon]\}.$$

We should notice that $|R_i - r_i|$ represents the deviation from the average return. $|R_i - r_i| x_i$ informs us how much that deviation impacts our portfolio via the weight of asset i in the portfolio, x_i. ϵ is a measure of risk if we think that larger deviations are less desirable than fewer deviations. For instance, we could link ϵ to a measure of average risk for the entire portfolio:

$$\epsilon = \frac{1}{N} \sum_{i=1}^{N} \sigma_i,$$

but this is just one possible way.

Note: The risk measure that we introduced penalizes assets in the portfolio whose returns exceeds the average return by more than a certain amount of market volatility. We could modify it to penalize

only assets that underperform. The only caveat is that, as we will see briefly, we need to be able to derive certain analytical results in order to create a suitable algorithm.

θ is a constant used to adjust the risk aversion. The smaller θ, the more conservative the investor.

We assume that the investor is still risk averse and rational, and therefore, their portfolio selection can be described by a bi-criteria optimization problem:

$$\max_{x_1,\ldots,x_N} \{\omega_p, \sum_{i=1}^{N} r_i x_i\}.$$

To make the notation more compelling, let

$$p(x_i) \equiv P[|R_i - r_i| x_i \leq \theta\epsilon], \;\; i = 1, \ldots, N.$$

Then, the maximization problem can be written as:

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$$\begin{aligned} &\max_{x_1,\ldots,x_N} \{\min p(x_i), \sum_{i=1}^{N} r_i x_i\} \\ &\text{s.t. } (x_1, \ldots, x_N) \in \mathcal{X}. \end{aligned}$$

which is the bi-criteria optimization problem. In this problem, we are still concerned about the volatility of returns, but it is quite different from the optimal problem we considered before, i.e.:

$$\begin{aligned} &\max_{x_1,\ldots,X_N} \sum_{i=1}^{N} r_i x_i - \frac{\theta}{2} \sum_{i=1}^{n} \sum_{j=1}^{N} \sigma_i \sigma_j x_i x_j \\ &\text{s.t. } (x_1, \ldots, x_N) \in \mathcal{X}. \end{aligned}$$

In the new setup, we are concerned about the probability of certain outcomes and not the overall variability.

This problem can be solved by introducing a new variable and N additional constraints:

$$\begin{aligned} &\max_{x_1,\ldots,x_N, y} \{y, \sum_{i=1}^{N} r_i x_i\} \\ &\text{s.t. } y \leq f(x_i)), \;\; i = 1, \ldots, N, \\ &(x_1, \ldots, x_N) \in \mathcal{X}. \end{aligned}$$

Using the assumption of normality, with some calculus, we find that

$$p(x_i) = 2 \int_0^{\frac{\theta\epsilon}{x_i}} \frac{1}{\sqrt{2\pi}\sigma_i} e^{-(x_i^2/2\sigma_i^2)} \, dz_i$$

where $z_i = R_i - r_i$. We are trying to minimize $f(x_i)$ and it should be easy to realize that $f(x_i)$ is monotonically decreasing with respect to x_i. Hence, since we need to have $y \leq f(x_i)$ and that to maximize our objective function above we need to push y to reach $\min_{i=1,\ldots,N} p(x_i)$, then for any number $y \in (0, 1)$ (recall that y is a probability!), the fact that $p(x_i)$ is monotonic implies that the upper bound for x_i is given by $p^{-1}(y)$. More specifically, we use the notation $\tilde{x}_i = p^{-1}(y)$ and

define $U_i = \min\{1, \tilde{x}_i\}$. Thus, for any fixed value of y, the optimization problem can be further re-written as a linear programming (LP) problem:

$$\begin{aligned} &\max_{x_1,\ldots,x_N} \sum_{i=1}^{N} r_i x_i \\ &\text{s.t.} \sum_{i=1}^{N} x_i = 1 \\ &\quad 0 \le x_i \le U_i, \;\; i = 1, \ldots, N. \end{aligned}$$

Now, assuming that none of the assets have the same returns and standard deviation (in continuous setting this probability is zero!), _Sun et al. _(see References) have shown that the problem can be solved as follows:

- Order the asset in order of return, such that $r_1 \ge r_2 \ge \ldots \ge r_N$.
- Find the integer $k \le N$ such that $\sum_{i=1}^{k-1} U_i < 1$, and $\sum_{i=1}^{k} U_i \ge 1$.
- Let

$$x_i^\star = \begin{cases} U_i & \text{if } i = 1, 2, \ldots, k-1; \\ 1 - \sum_{i=1}^{k-1} U_i & \text{if } j = k, \\ 0 & \text{otherwise.} \end{cases}$$

and $x_1^\star, x_2^\star, \ldots, x_N^\star$ is the optimal solution for our problem.

Note that in the previous solution, we fixed y. This is not different than what we did when computing the efficient frontier in the sense of Markowitz's mean-variance optimization where we can fix σ and then find the optimum portfolio and its return, and we keep repeating the process.

Another interesting type of risk function is the group known as ℓ functions. In practice, the ones used are ℓ_1, ℓ_2, and ℓ_∞. The name comes from their respective metrics in analysis and functional analysis. The ℓ_2 risk measure is just the classic M-V formulation and it amounts to quadratic programming, and therefore, there is not much to add. The ℓ_1 risk measure can be introduced like this:

$$\begin{aligned} &\min_{x_1,\ldots,x_N} L(x_1, \ldots, x_N) \\ &\text{s.t.} \sum_{i=1}^{N} r_i x_i \ge \rho, \\ &\quad \sum_{i=1}^{N} x_i = 1, \\ &\quad 0 \le x_i \le u_i, \;\; i = 1, 2, \ldots, N \end{aligned}$$

where (see the paper by Konno and Suzuki listed among the references)

$$L(x_1, \ldots, x_N) \equiv E\left[\left|\sum_{i=1}^{N} R_i x_i - E[\sum_{i=1}^{N} R_i x_i]\right|\right] = \sqrt{\frac{2}{\pi}}\sigma(x_1, \ldots, x_N)$$

with

$$\sigma(x_1,\ldots,x_N) \equiv \sqrt{\sum_{i=1}^{N}\sum_{j=1}^{N}\sigma_{ij}x_i x_j}.$$

The interesting part to notice is that this is now an LP problem! Then, the risk function can be approximated like this:

$$L(x_1,\ldots,x_N) = \frac{1}{T}\sum_{t=1}^{T}\left|\sum_{i=1}^{N}(r_i t - r_i)x_i\right|$$

and letting $a_{it} = r_{it} - r_i$ for each $i = 1, 2, \ldots, N$ and $t = 1, 2, \ldots, T,$ we can write a new LP problem as:

$$\begin{aligned}
&\min_{x_1,\ldots,x_N} \frac{1}{T}\sum_{i=1}^{T} y_t \\
&\text{s. t. } y_t + \sum_{i=1}^{N} a_{it}x_i \geq 0, \;\; t_1,\ldots,T, \\
&\quad y_t - \sum_{i=1}^{N} a_{it}x_i \geq 0, \;\; t_1,\ldots,T, \\
&\quad \sum_{i=1}^{N} r_i x_i \geq \rho, \\
&\quad \sum_{i=1}^{N} x_i = 1, \\
&\quad 0 \leq x_i \leq u_i, \;\; i = 1,\ldots,N.
\end{aligned}$$

In this setup,u_i is the maximum percentage of capital that can be invested in the i-th asset. In addition, while the original problem has N variables, the new one has $N + T.$

The ℓ_∞ formulation of the problem looks like this:

$$\begin{aligned}
&\min\left(\max_{x_1,\ldots,x_N} q_i r_i, -\sum_{i=1}^{N} r_i x_i\right) \\
&\text{s.t. } (x_1,\ldots,x_N) \in \mathcal{X}
\end{aligned}$$

where The problem can then be transformed to:

$$\begin{aligned}
&\min_{x_1,\ldots,X_N,y} \lambda y + (1-\lambda)\left(-\sum_{i=1}^{N} r_i x_i\right) \\
&\text{s.t. } q_i x_i \leq y, \;\; i = 1,\ldots,N \\
&\quad (x_1,\ldots,x_N) \in \mathcal{X}
\end{aligned}$$

for a fixed $\lambda \in (0,1).$ The equivalence in this case is not straightforward, and interested students can find some references at the end of the lesson. Using the latest program, it is possible to show that for any $\lambda \in (0,1),$

$$x_i^\star = \begin{cases} \frac{W_0}{q_i}\left(\sum_{j\in H(\lambda)} \frac{1}{q_j}\right)^{-1} & \text{if } i \in H(\lambda); \\ 0 & \text{otherwise}, \end{cases} \quad y^\star = W_0\left(\sum_{j\in H(\lambda)} \frac{1}{q_j}\right)^{-1}$$

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with $H(\lambda)$ defining the set of assets to be invested and that is determined according to a complicated rule provided in the Cai et al.'s paper.
The different approaches, as expected, return different efficient frontiers, and it is up to the practitioner to decide which one is the most suitable for specific applications.

3. An Application of the Probabilistic Approach for Optimal Portfolio Selection

So far we have covered the main ideas of PSO. PSO is not a small feat to pull off, and we discussed some of the main issues. Then, we considered probabilistic measures to determine optimal portfolios because the mean-variance setup is not ideal in all cases.

The following **required** reading:

Probabilistic approach for optimal portfolio selection using a hybrid Monte Carlo simulation and Markowitz model

is reasonably simple and illustrates well some of the nuances implicit in probabilistic approaches to optimal portfolio determination and, because of the simulations, can also be linked to a simple instance of PSO.

4. Conclusion

In this lesson, we have covered the basics of probability scenario optimization and looked at alternative ways to deal with risk in portfolio problems. Variance is just one way to do it. Finally, the application contained in the required reading illustrates some the practical nuances that practitioners need to deal with in solving real problems.

The next lesson will introduce a totally different topic as we move to consider behavioral finance, behavioral portfolio theory, and how accounting for individuals' emotional and cognitive biases changes the way we look at the problem of portfolio optimization.

Pyportfolioopt Package

We are at the end of Module 3 and by now the key traditional approaches to portfolio optimization have been considered. Furthermore, at this point in time, students should have learned several algorithms that are necessary to implement those techniques. Thus, it is the right time to bring up that there exist several large packages that allow practitioners to implement the methodologies that were considered and more. One such package is **pyportfolioopt**.

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